Two Clustering Algorithms

K-Means and DBSCAN

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What are they?

K-Means:

- Partitions the data into k clusters by minimizing distance to cluster centroids.
- Works best for spherical, evenly sized clusters.
- Requires you to choose k (often with the Elbow method).
- Fast and widely used, but sensitive to initial centroid placement

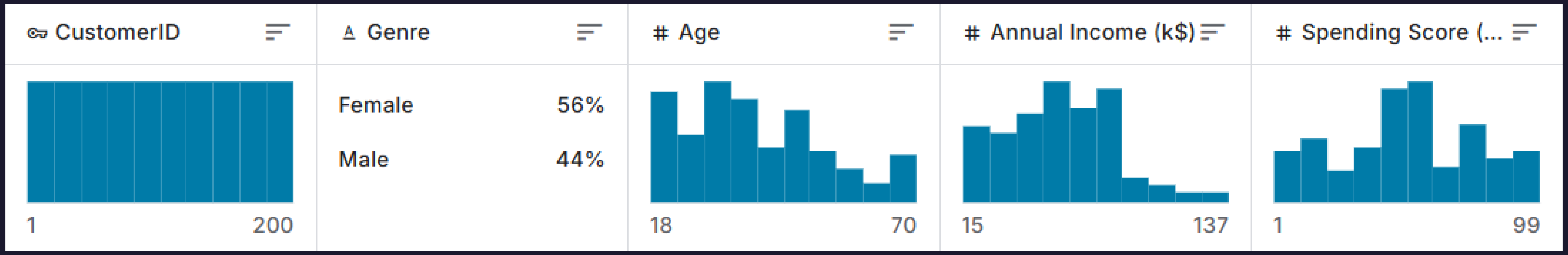
DBSCAN:

- Groups points that are closely packed together (within eps distance, with at least min_samples neighbors).
- Does not require pre-setting the number of clusters.
- Can find arbitrarily shaped clusters and identify noise/outliers.
- Works best when clusters are of similar density; sensitive to choice of eps and min_samples.

Dataset

Kaggle's Mall Customer dataset (https://www.kaggle.com/datasets/shwetabh123/mall-customers)

- 200 data vectors
- 5 values – ID, Gender, Age, Annual Income (in $1000's), Spending Score (1-100)
  - ID: Just numerical key index value
  - Gender: Binary categorical data point: Male (44%) or Female (56%) (Listed as 'Genre' in .csv file)
  - Age: Age of each entry, in years: youngest at 18, oldest at 70
  - Annual Income: Income value of each entry, in thousands of dollars: $15k-$137k
  - Spending Score: rating of how much each entry spends in this mall from 1-100, with 100 the highest spenders



Preprocessing

- Used the 3 numerical columns per vector – Age, Income, and Spending
- Retained Gender information for post-clustering graphing information
- Normalized all three columns using Z-score normalization:

$$Z = (X - \mu) / \sigma$$

  where 'X' is the original value, 'μ' is the mean of the dataset, and 'σ' is the standard deviation

Bridges2 Execution

```
(clustering) [schoolcr@bridges2-login014 schoolcr]$ interact -gpu -t 10

A command prompt will appear when your session begins
"Ctrl+d" or "exit" will end your session

--partition=GPU-small,GPU-shared --gpus=v100:1 --time=10
salloc -J Interact --partition=GPU-small,GPU-shared --gpus=v100:1 --time=10
salloc: Pending job allocation 34931313
salloc: job 34931313 queued and waiting for resources
salloc: job 34931313 has been allocated resources
salloc: Granted job allocation 34931313
salloc: Waiting for resource configuration
salloc: Nodes v001 are ready for job
(base) [schoolcr@v001 schoolcr]$ conda activate clustering
(clustering) [schoolcr@v001 schoolcr]$ python clustering.py
Chosen k (elbow): 5
Saved interactive plot to kmeans_3d_clusters_interactive.html
K-Means silhouette score: 0.41664341513732767
DBSCAN final: eps=0.634, min_samples=6, clusters=5, noise_pts=30
DBSCAN DBI: 1.1247621417987184 CHI: 34.27778408935683 5
Saved dbscan_k_distance_knee.png
(clustering) [schoolcr@v001 schoolcr]$
```

Comparing Methods: Strengths, Weaknesses, Parameters, and Metrics

Clustering Algorithm	Strengths	Weaknesses	Parameters	Metrics
K-Means	• Clean clusters • Interpretable • Efficient	• Sensitive to initial centroid location • Cannot identify outliers	K (Number of clusters): 5	Silhouette score (measure of similarity to own cluster vs. other clusters): 0.42
DBSCAN	• Find arbitrary shapes • Identified outliers/noise	• Very sensitive to epsilon value • Weaker cohesion	Epsilon (Max distance between points): 0.634 Min_samples (Min points within epsilon to start a cluster): 6	DBI (intra-cluster distance vs inter-cluster distance): 1.12 CHI (between-cluster dispersion vs within-cluster dispersion): 34.3

Comparing Methods: Preprocessing

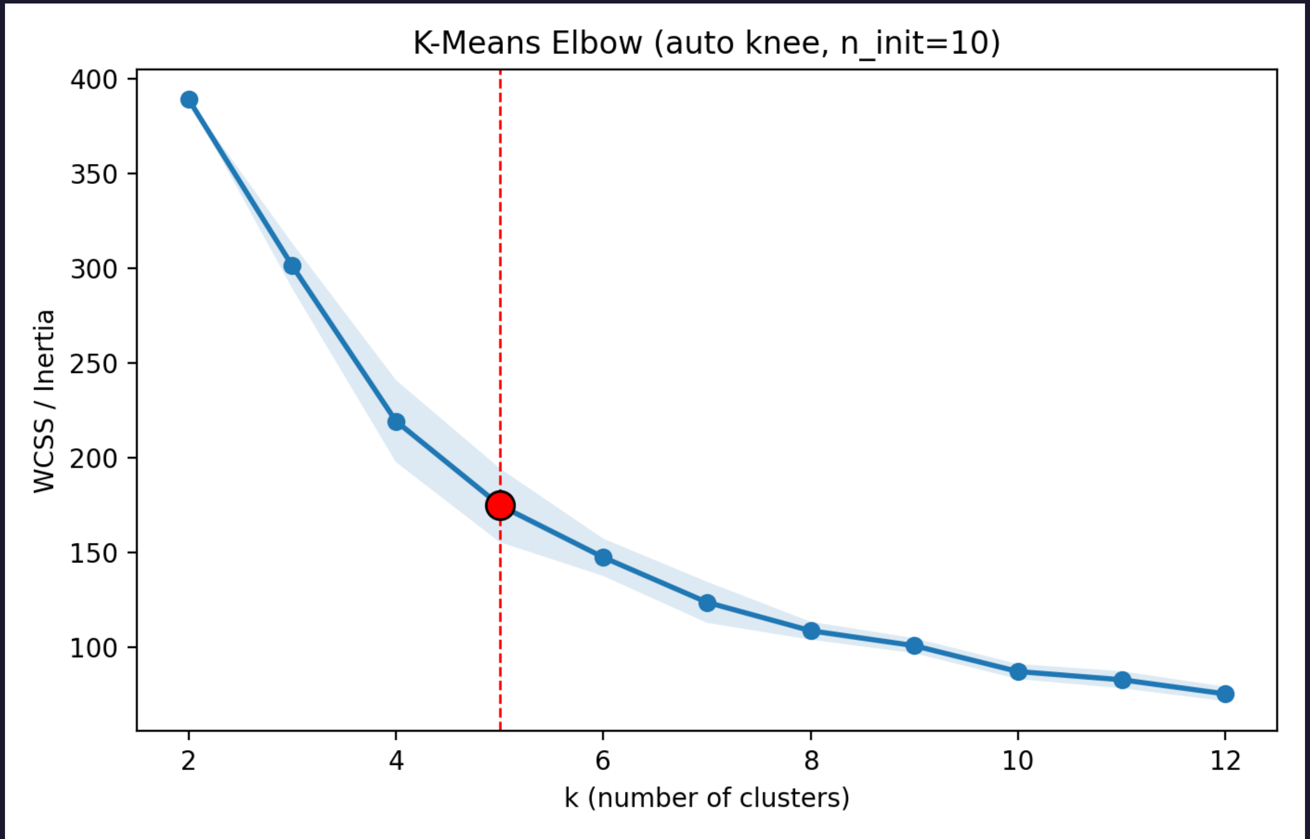
K-Means:

- Used Elbow Method to determine ‘k’
- Best results at k=5
- All distinct clusters when graphed in 3D

DBSCAN:

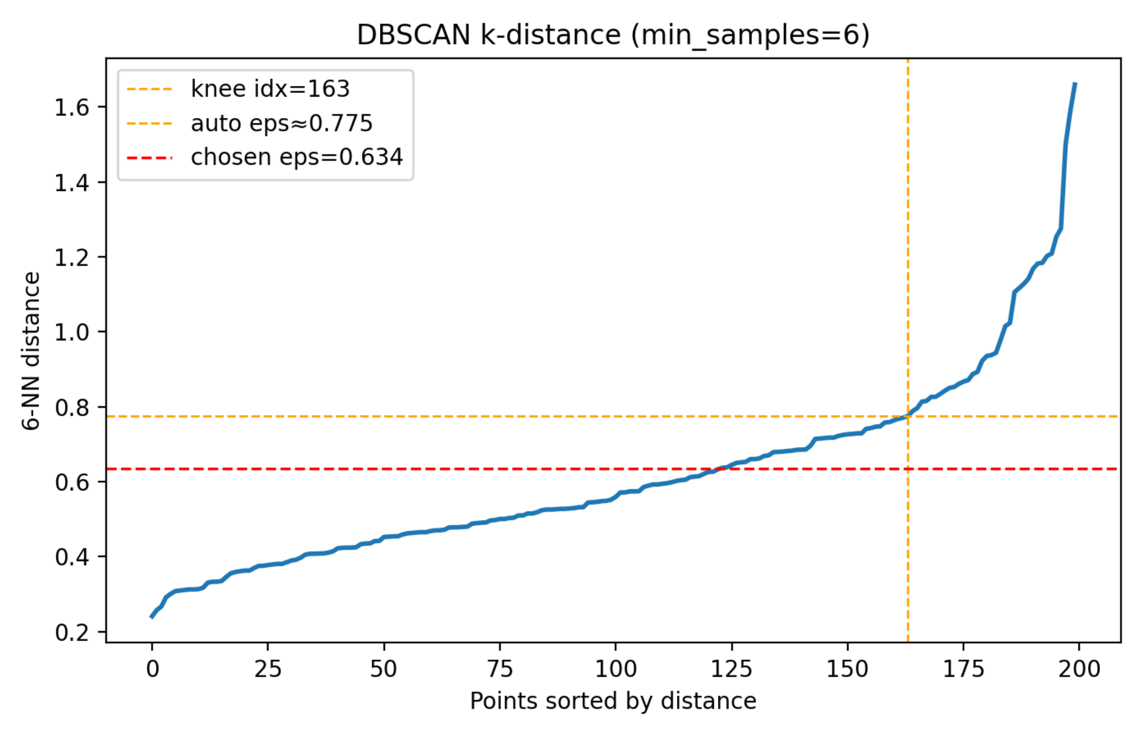
- Used Elbow Method to determine initial epsilon/min samples combo
- Then conducted a sweep of similar values to determine best results
- Eps = 0.634, min samples = 6
- Left a small amount (~7.5%) of noise values (non-clustered), and one very small cluster.

K-Means Elbow Plot

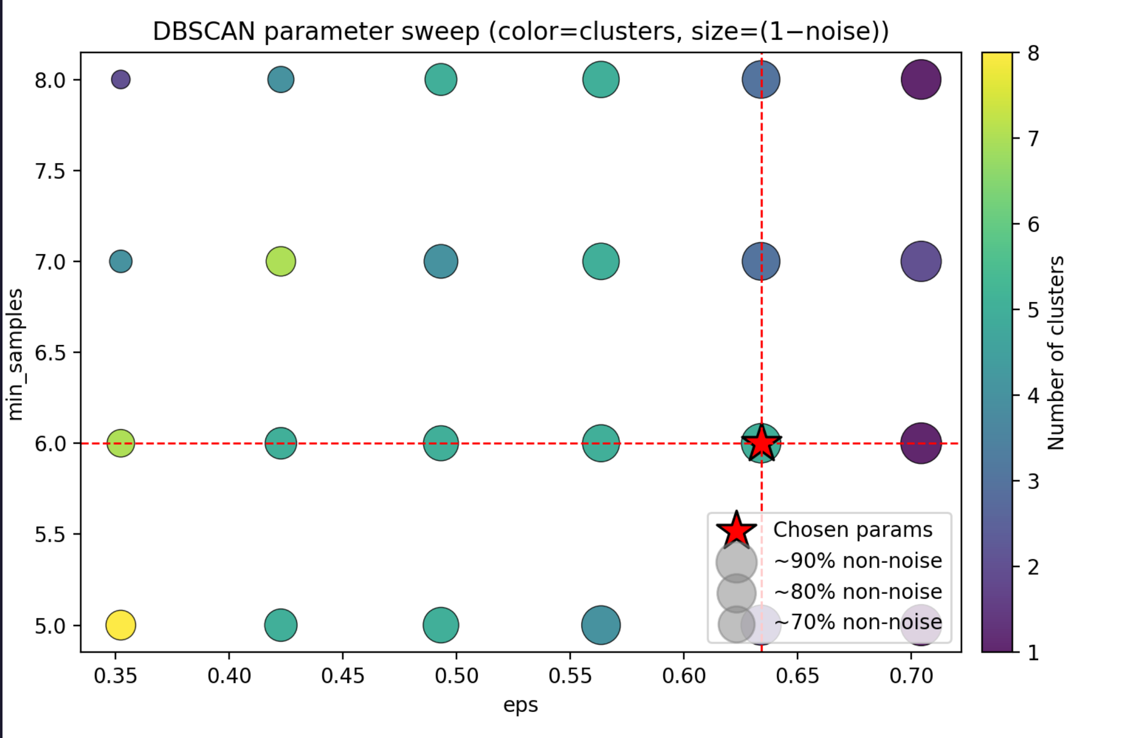


DBSCAN Parameter Selections

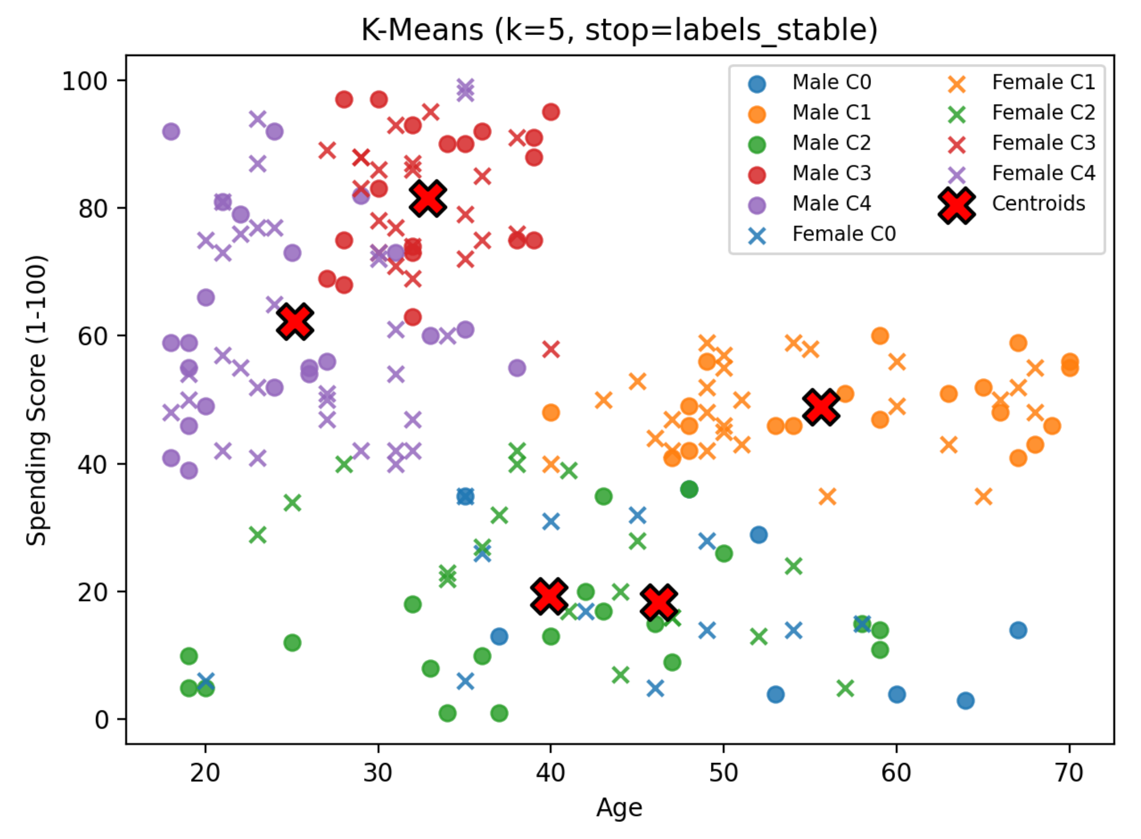
DBSCAN KNEE PLOT

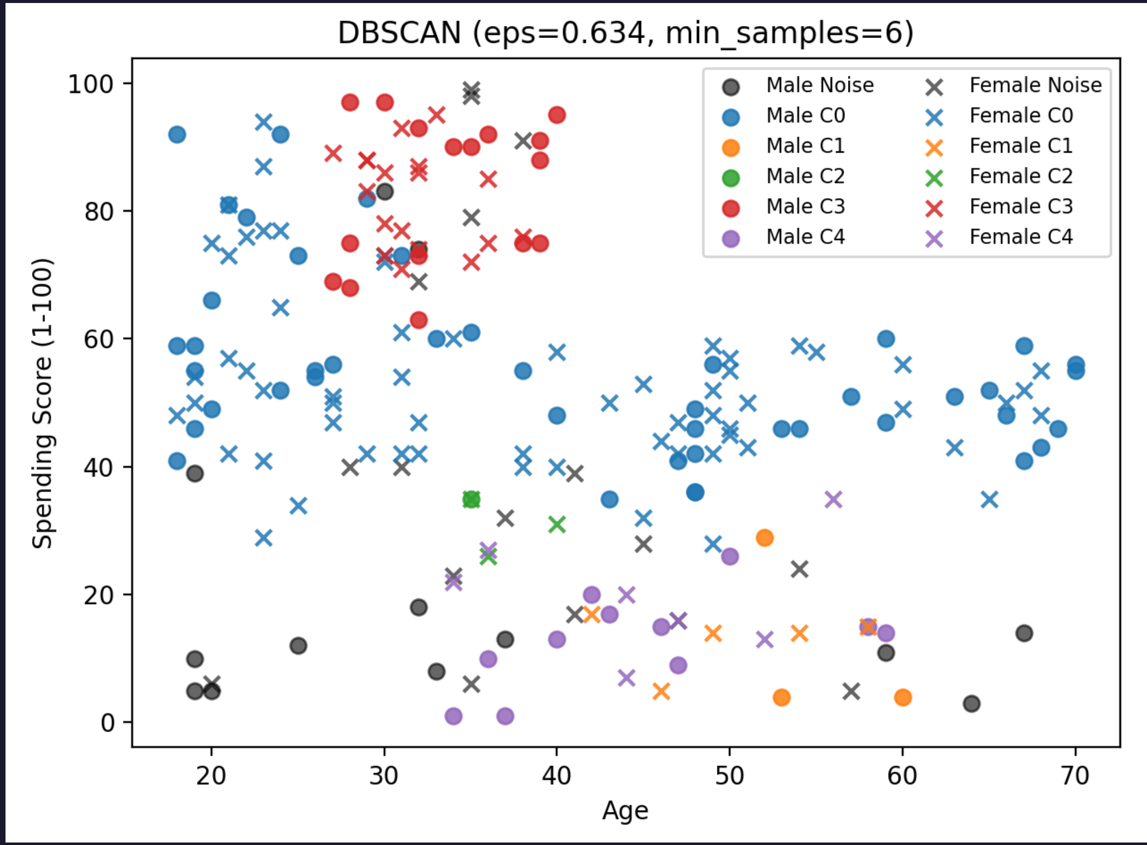


DBSCAN SWEEP RESULTS

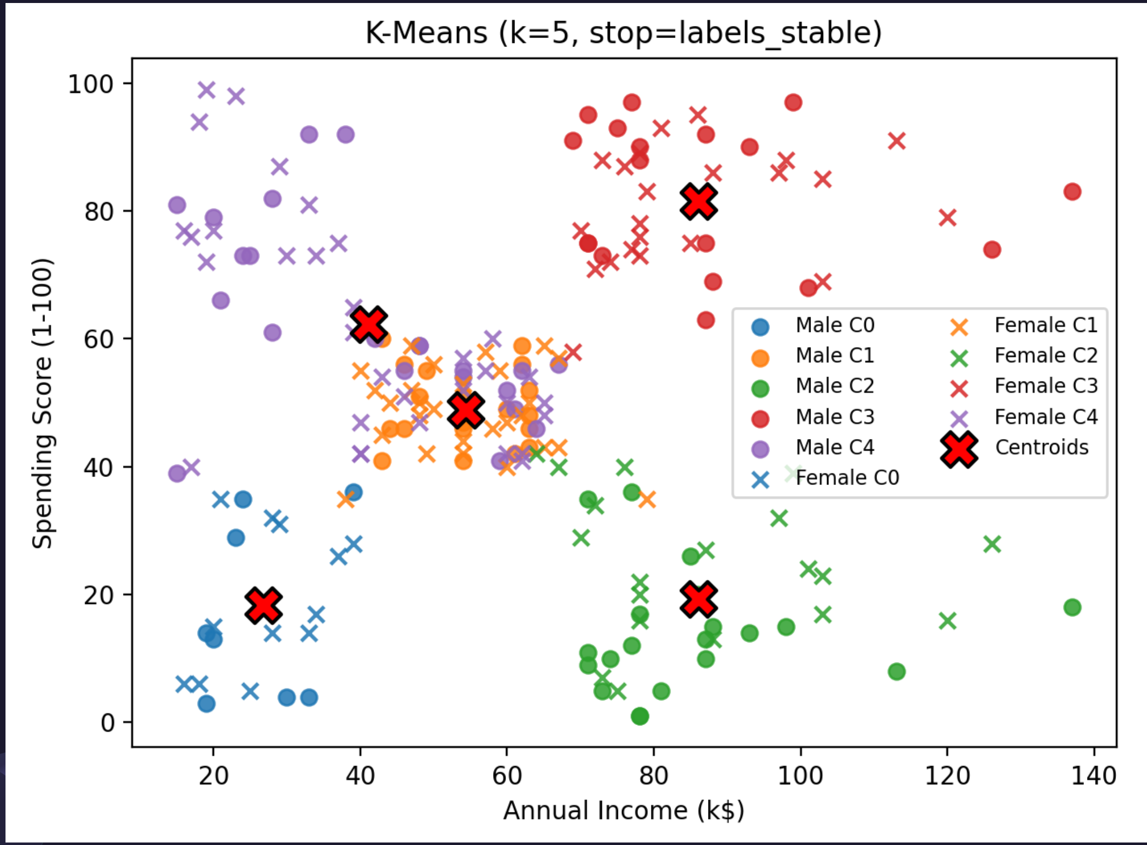


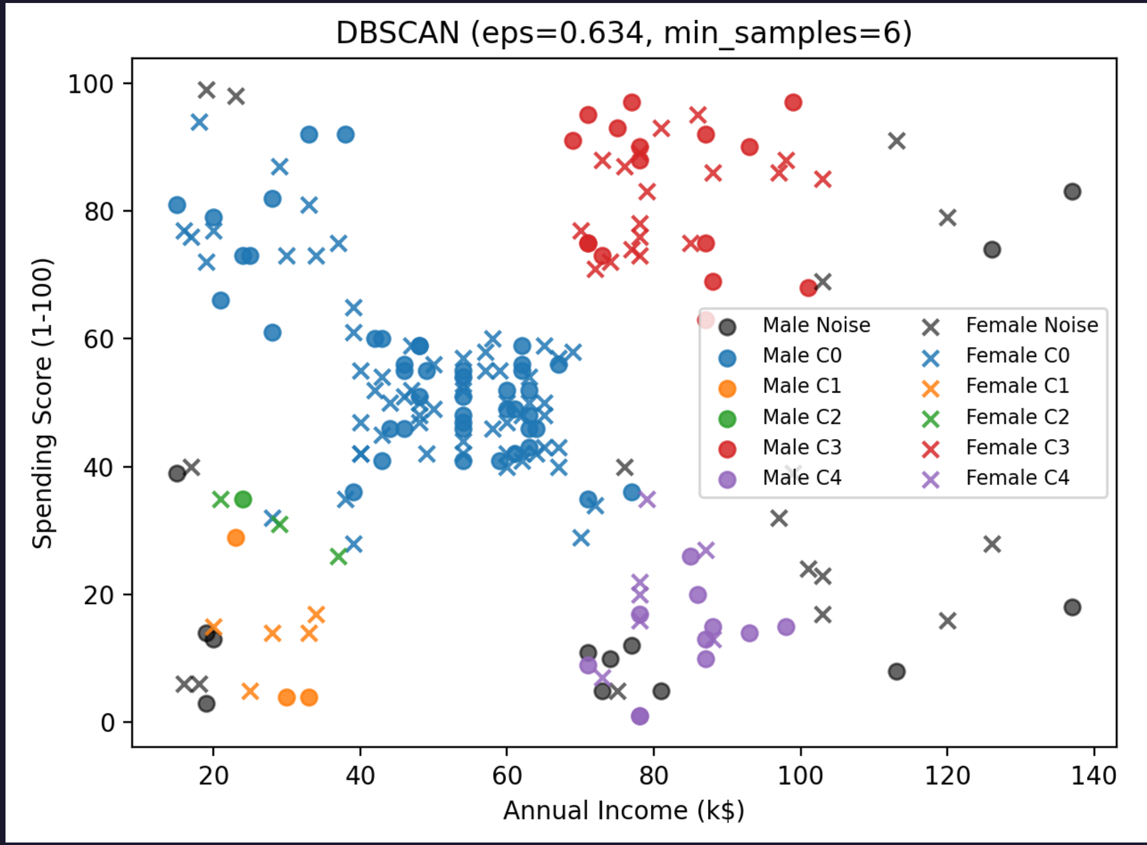
Comparing Scatterplots: Age vs Score



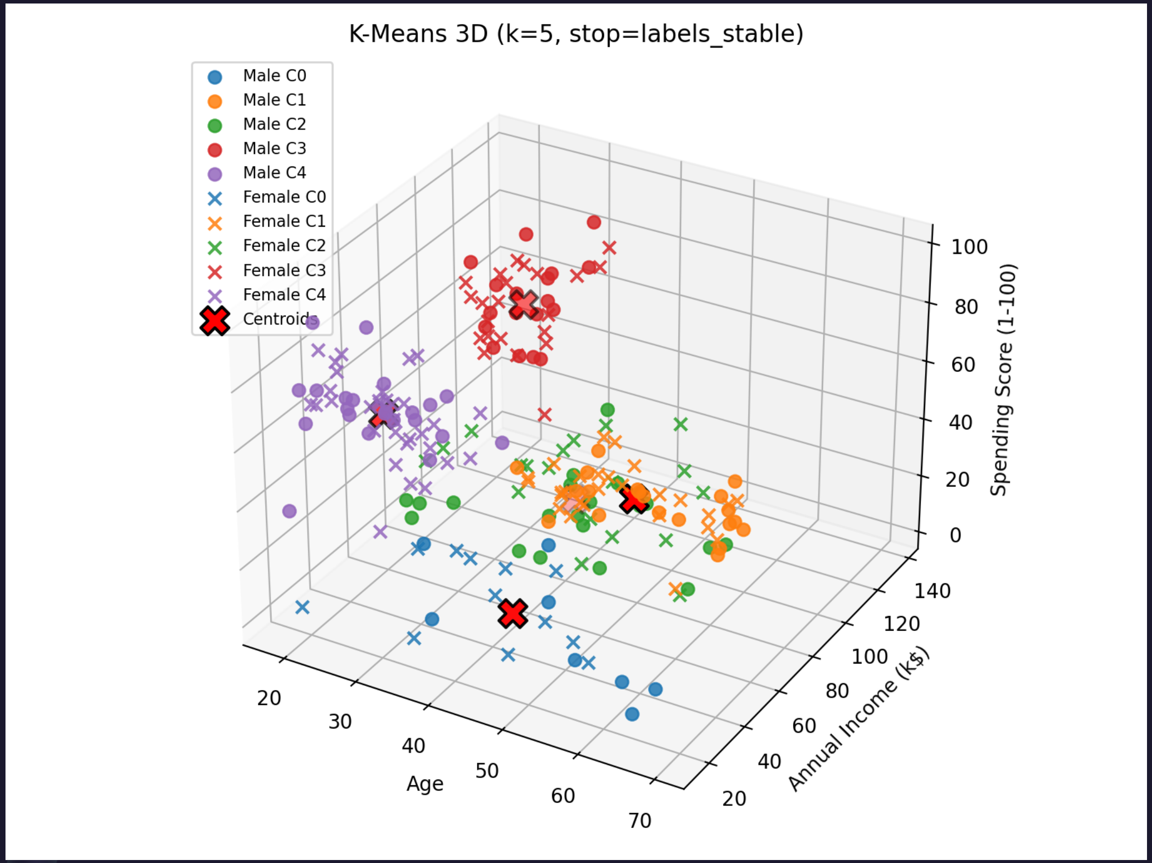


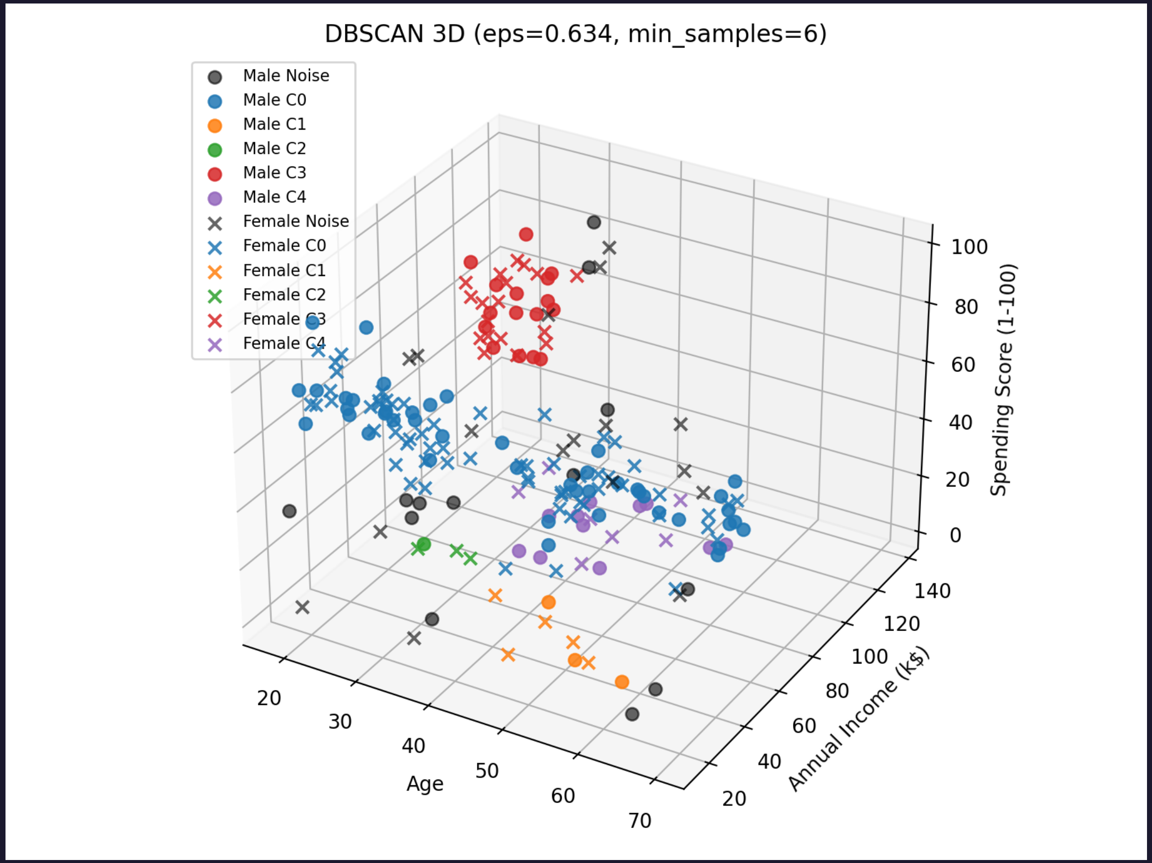
Comparing Scatterplots: Income vs Score





Comparing Results in 3-D





K-Means Results

Output evaluated by Silhouette Score: ~0.42, indicates decent separation and cohesion, thus meaningful and actionable, but there is some overlap, which is expected in real-world data.

- Younger low-income shoppers tend to be some of the largest spenders, with an interesting negative correlation with income
- A group of low-income shoppers across all age groups that do not spend much money
- Younger high-income shoppers tend to also have the highest spending scores
- There is an interesting income gap of the highest scored shoppers between about $40k and $70k, where regardless of age, they tend to be in the middle ground of spending score, ~ 40-60.
- Outside of a few outliers, the older wealthy group tended to have the lowest scores.

DBSCAN Results

30 noise points, ~7.5% are outliers. Evaluated using Davies-Bouldin Index: 1.12 (indicates clusters are moderately well separated) and Calinski-Harabasz Index: 34.3 (shows presence of structure). Overall, strong clustering, however, not quite as tight as K-Means

- DBSCAN also identified 5 clusters, and for the most part, they are very similar to the K-Means, but there are a few differences.
- The middle-of-the-road group all were clustered with the high-spending, low-income youth.
- There is a small cluster of just 4 elements at the low end of the income scale, all roughly 30-40 years old, and all scored roughly 25-35.
- There are also several outliers identified, some as mentioned with the K-Means clustering, but basically all around the edges there are some noise outliers that did not cluster with any group.
- Unlike K-Means, DBSCAN allowed for clusters of different densities, grouping together some customers that K-Means had split, while splitting others that K-Means had clustered.

Business Insights

For the young, high-spending, low-income group, (K-Means cluster 1) perhaps offer a variety of loyalty and/or specialty-classification (such as student or military) discounts to encourage more shopping.

For the median group, with income roughly in the $40k-$65k range, mall-wide promotions, packaged/bundled deals, and membership perks could help drive further sales.

For the wealthy low-spenders (cluster 4 in both methods), more exclusive and/or luxury stores and experiences could draw them in for additional shopping

Contributions

- Alexander: Algorithm implementation, DBSCAN sweep tuning, remote execution
- Robert: Data preprocessing, elbow/knee method plots
- Richard: Visualization cleanup, presentation prep